Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.38_k1)

Metric

$$ds^2 = -\mathrm{d}t \otimes \mathrm{d}t + \left(-\frac{\left(\frac{\partial Y}{\partial r}\right)^2}{\epsilon f^2 - 1}\right) \mathrm{d}r \otimes \mathrm{d}r + Y^2 \mathrm{d}\theta \otimes \mathrm{d}\theta + Y^2 \sin(\theta)^2 \, \mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$(x^\mu) = (t, r, \theta, \phi)$$

Metric Functions

$$f(r) = f$$

$$Y(r, t) = Y$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}}\right) dt + \left(\sqrt{\frac{1}{2}} \sqrt{-\frac{1}{\epsilon f^2 - 1}} \frac{\partial}{\partial r} Y\right) dr$$

$$n = \left(\sqrt{\frac{1}{2}}\right) dt + \left(-\sqrt{\frac{1}{2}} \sqrt{-\frac{1}{\epsilon f^2 - 1}} \frac{\partial}{\partial r} Y\right) dr$$

$$m = \left(\frac{1}{2} \sqrt{2} Y\right) d\theta + \left(\frac{1}{2} i \sqrt{2} Y \sin(\theta)\right) d\phi$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}Y\right) d\theta + \left(-\frac{1}{2}i\sqrt{2}Y\sin\left(\theta\right)\right) d\phi$$

Contravariant components

$$l = \left(-\frac{1}{2}\sqrt{2}\right) dt + \left(\frac{\sqrt{2}\sqrt{-\epsilon f^2+1}}{2\frac{\partial}{\partial r}Y}\right) dr$$

$$n = \left(-\frac{1}{2}\sqrt{2}\right) dt + \left(\frac{\sqrt{2}\epsilon f^2}{2\sqrt{-\epsilon f^2+1}\frac{\partial}{\partial r}Y} - \frac{\sqrt{2}}{2\sqrt{-\epsilon f^2+1}\frac{\partial}{\partial r}Y}\right) dr$$

$$m = \left(\frac{\sqrt{2}}{2Y}\right) d\theta + \left(\frac{i\sqrt{2}}{2Y\sin\left(\theta\right)}\right) d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2Y}\right) d\theta + \left(-\frac{i\sqrt{2}}{2Y\sin\left(\theta\right)}\right) d\phi$$

Directional Derivatives

$$D = -\frac{1}{2}\sqrt{2}\,\partial_t + \frac{\sqrt{2}\sqrt{-\epsilon f^2+1}}{2\frac{\partial}{\partial r}Y}\,\partial_r$$

$$\Delta = -\frac{1}{2}\sqrt{2}\,\partial_t + \frac{\sqrt{2}\epsilon f^2}{2\sqrt{-\epsilon f^2+1}\frac{\partial}{\partial r}Y} - \frac{\sqrt{2}}{2\sqrt{-\epsilon f^2+1}\frac{\partial}{\partial r}Y}\,\partial_r$$

$$\delta = \frac{\sqrt{2}}{2Y}\,\partial_\theta + \frac{i\sqrt{2}}{2Y\sin\left(\theta\right)}\,\partial_\phi$$

$$\overline{\delta} = \frac{\sqrt{2}}{2Y}\,\partial_\theta - \frac{i\sqrt{2}}{2Y\sin\left(\theta\right)}\,\partial_\phi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}\epsilon f\left(r\right)^{2}}{2\sqrt{-\epsilon f\left(r\right)^{2}+1}Y} + \frac{\sqrt{2}\frac{\partial}{\partial t}Y}{2\,Y} - \frac{\sqrt{2}}{2\sqrt{-\epsilon f\left(r\right)^{2}+1}Y}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{\sqrt{2}\epsilon f\left(r\right)^{2}}{2\sqrt{-\epsilon f\left(r\right)^{2}+1}Y} - \frac{\sqrt{2}\frac{\partial}{\partial t}Y}{2\,Y} - \frac{\sqrt{2}}{2\sqrt{-\epsilon f\left(r\right)^{2}+1}Y}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\cos\left(\theta\right)}{4\,Y\sin\left(\theta\right)}$$

$$\beta = \frac{\sqrt{2}\cos\left(\theta\right)}{4\,Y\sin\left(\theta\right)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2}\frac{\partial^{2}}{\partial r\partial t}Y}{4\,\frac{\partial}{\partial r}Y}$$

$$\gamma = \frac{\sqrt{2}\frac{\partial^{2}}{\partial r\partial t} Y}{4 \frac{\partial}{\partial r} Y}$$

Ricci Tensor Components

$$\Phi_{00} = \frac{\epsilon f\left(r\right) \frac{\partial}{\partial r} f\left(r\right)}{2 Y \frac{\partial}{\partial r} Y} + \frac{\frac{\partial^{2}}{\partial r\partial t} Y \frac{\partial}{\partial t} Y}{2 Y \frac{\partial}{\partial r} Y} - \frac{\frac{\partial^{2}}{(\partial t)^{2}} Y}{2 Y}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{\epsilon f\left(r\right)^{2}}{4 Y^{2}} - \frac{\frac{\partial^{3}}{\partial r(\partial t)^{2}} Y}{4 \frac{\partial}{\partial r} Y} + \frac{\frac{\partial}{\partial t} Y^{2}}{4 Y^{2}}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{\epsilon f\left(r\right) \frac{\partial}{\partial r} f\left(r\right)}{2 Y \frac{\partial}{\partial r} Y} + \frac{\frac{\partial^{2}}{\partial r\partial t} Y \frac{\partial}{\partial t} Y}{2 Y \frac{\partial}{\partial r} Y} - \frac{\frac{\partial^{2}}{(\partial t)^{2}} Y}{2 Y}$$

$$\Lambda = \frac{\epsilon f\left(r\right)^{2}}{12 Y^{2}} + \frac{\epsilon f\left(r\right) \frac{\partial}{\partial r} f\left(r\right)}{6 Y \frac{\partial}{\partial r} Y} + \frac{\frac{\partial^{3}}{\partial r(\partial t)^{2}} Y}{12 \frac{\partial}{\partial r} Y} + \frac{\frac{\partial^{2}}{\partial r\partial t} Y \frac{\partial}{\partial t} Y}{6 Y \frac{\partial}{\partial r} Y} + \frac{\frac{\partial}{\partial t} Y^{2}}{12 Y^{2}} + \frac{\frac{\partial^{2}}{(\partial t)^{2}} Y}{6 Y}$$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$$\Psi_2 = -\frac{\epsilon f\left(r\right)^2}{6\,Y^2} + \frac{\epsilon f\left(r\right)\frac{\partial}{\partial r}f\left(r\right)}{6\,Y\frac{\partial}{\partial r}Y} - \frac{\frac{\partial^3}{\partial r(\partial t)^2}Y}{6\,\frac{\partial}{\partial r}Y} + \frac{\frac{\partial^2}{\partial r\partial t}Y\,\frac{\partial}{\partial t}Y}{6\,Y\frac{\partial}{\partial r}Y} - \frac{\frac{\partial}{\partial t}Y^2}{6\,Y^2} + \frac{\frac{\partial^2}{(\partial t)^2}Y}{6\,Y}$$

$\Psi_3 = 0$

$\Psi_4 = 0$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"